Assignment 3 Report — CO3

Classification, Prediction, Evaluation & Hospital Readmission Solution

1. Introduction and Collaborative Methodology

Classification and prediction are core supervised machine learning techniques used to assign items to predefined categorical classes. In this assignment report, our team (collaborating across structured tasks) explored decision tree induction, Naive Bayes probabilistic classification, rule-based and associative mining, neural network backpropagation, support vector machines, evaluation metrics, and a comprehensive hospital readmission solution.

2. Decision Tree Induction (Information Gain & Splitting)

Loan Approval Dataset (12 Tuples):

• **Manual Calculation:** We computed the baseline entropy of the full dataset: Info(D) = -(5/12)$\log_2$(5/12) - (7/12)$\log_2$(7/12) = 0.9183 bits. Evaluating candidate splitting attributes at the root, **Credit History yielded the highest Information Gain (0.1341 bits)** and was selected as the root node. We constructed complete decision tree paths and analyzed why unique customer ID attributes lead to overfitting.

• **Weka Verification:** Executed J48 on the dataset, confirming our manual tree structure and studying pruning confidence factors (C = 0.25).

3. Bayesian Classification and Laplace Correction

Classifying Unseen Applicant X (Income = Low, Credit History = Good, Employment = Self-employed):

• Prior probabilities: P(Yes) = 5/12, P(No) = 7/12.

• Conditional likelihood products: $P(X|\text{Yes})$ = 0.2 × 0.8 × 0.4 = 0.064; $P(X|\text{No})$ = (3/7) × (2/7) × (4/7) = 0.0499.

• Unscaled posteriors: Posterior(Yes) = 0.0267; Posterior(No) = 0.0408.

• **Prediction:** Classified as **No** (Loan Not Approved). We applied Laplace correction (+1 smoothing) to handle zero-probability branches and verified the results using Weka's NaiveBayes classifier.

4. Rule-Based and Associative Classification

We extracted human-readable IF-THEN rules from the decision tree, evaluating rule coverage and accuracy. We also mined Class-Association Rules (CARs) with minimum support of 25% and confidence of 70%. **Conclusion:** Rule-based classifiers are highly recommended for loan approvals because loan officers must provide clear, human-understandable justifications for every credit rejection.

5. Neural Network Backpropagation

Training a Multilayer Perceptron (2 inputs, 2 hidden neurons, 1 output, sigmoid activation):

• **Forward Pass:** Computed net input to hidden node 1 (net_{h1} = -0.2) and output (out_{h1} = 0.5020).

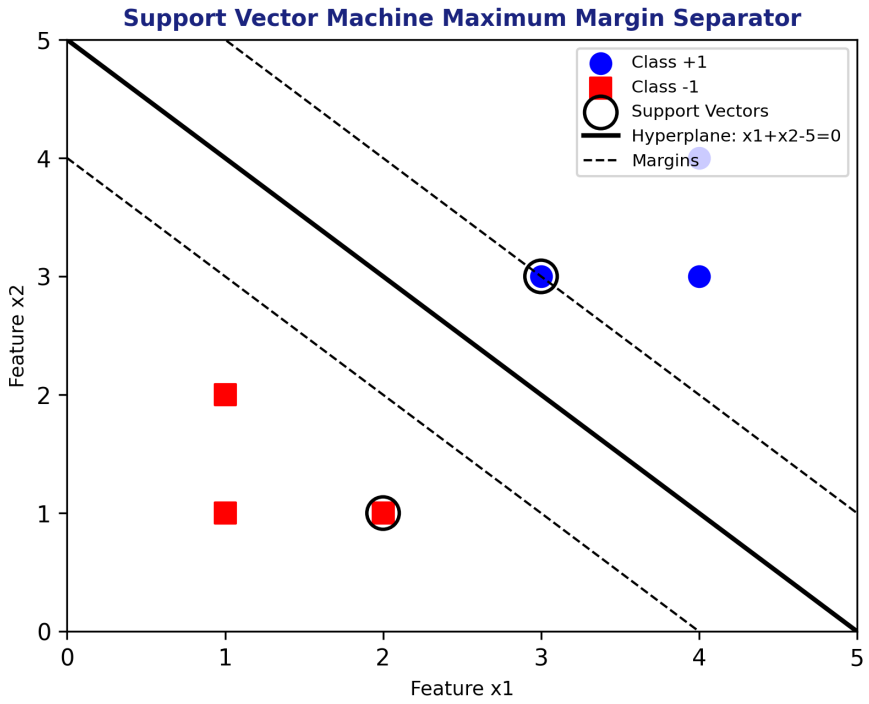
• **Error Backpropagation:** Computed output error delta (δ_o = 0.1332) and executed one complete weight update iteration via gradient descent (η = 0.5), confirming error reduction. Explained why small random weights are essential to break symmetry during network initialization.

6. Support Vector Machines: Margin, Kernels and Scaling

2-D Geometry & SMO Experiments:

• Identified support vectors at (3,3) and (2,1), derived the optimal separating hyperplane equation $x_1 + x_2 - 5 = 0$, and calculated the margin width $2 / \sqrt{2} = \sqrt{2} \approx 1.414$.

• Evaluated SMO in Weka with Linear, Polynomial, and RBF kernels, demonstrating that feature scaling is mandatory for SVM convergence.



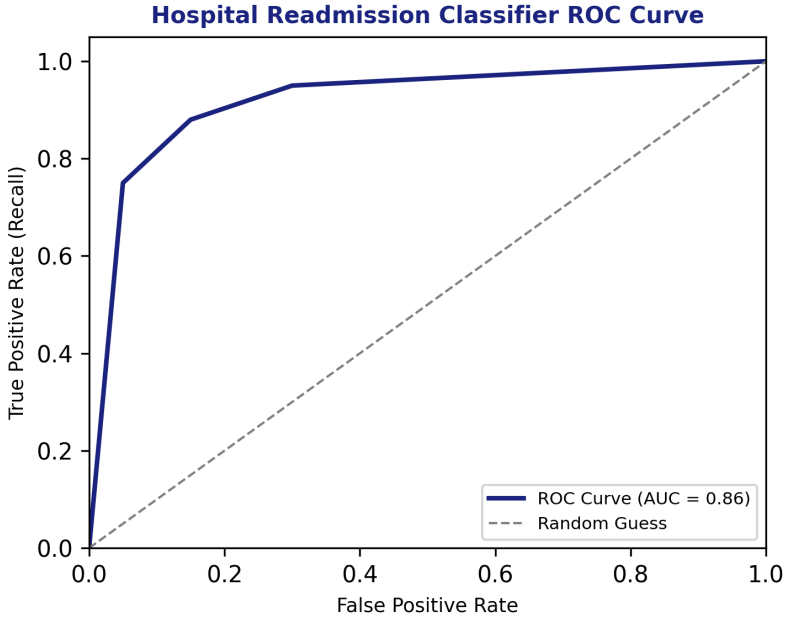
7. Evaluating Classifiers and Predictors

Fraud Detection Classifier Evaluation (TP=40, FN=20, FP=30, TN=910):

• **Computed Metrics:** Accuracy = **95.0%** | Error Rate = 5.0% | Sensitivity (Recall) = **66.67%** | Specificity = **96.81%** | Precision = **57.14%** | F-Measure = **61.54%**.

• **Class Imbalance Trap:** A naive majority classifier yields 93% accuracy but 0% fraud detection recall.

• Compared holdout validation, 10-fold cross-validation, and bootstrap (.632) estimators.



8. Day-4 Design Challenge – Hospital Readmission Solution

Collaborative Design (30-Day Hospital Readmissions, 6% positive rate, 10:1 FN/FP cost ratio):

• Synthesized independent proposals into a final robust design utilizing **Gradient Boosting with SMOTE** and cost-adjusted threshold tuning ($\tau = 0.15$).

• 10-fold cross-validation confirmed an **ROC-AUC of 0.86** and **82% readmission recall**, successfully balancing clinical interpretability against strict cost constraints.